Lab Course:
Projects in Recommender Systems

Kick-off Presentation Summer 2026

Chair of Connected Mobility
Munich, April 20, 2026

Lab Course Overview

- Practical course module IN0012 or IN2106
  - Conducted online via Teams (no in-person meetings, no recordings)

- 2 groups (different advisors) with teams of 2 students each
  - Team assignment completed, not possible to fulfill all preferences
  - All topics are somewhat flexible in terms of what exactly to do, and which technology to use etc

- Schedule
  - Three milestone meetings (~1.5 hours), each group separately
  - Mon 06.07.2026 (tentative), 14:00 (3+ hours): Final presentations (all students, both groups) -> can we start earlier ?
  - Fri 17.07.2026: Deadline to complete and submit final documentation

Projects: [A1] RAG for Tourism Recommender Systems

Proposed Work

1. Improving Existing Systems
   - Enhance the RAG model from our "RecSoGood" paper (referenced below) by addressing bottlenecks in retrieval and generation components.
   - Integrate domain-specific knowledge for better recommendations.
2. Synthetic Data Generation
   - Develop synthetic datasets simulating user preferences and diverse travel contexts.
   - Improve model generalization and robustness through tailored data

Feel free to suggest more options, ideas in this direction.

Required (Primary) Skills

Python, understanding of LLMs, data analysis, machine learning or similar

Projects: [A2] Gamification in Sustainable TRS

Proposed Work

1. Develop an application that uses gamification techniques to encourage users to make sustainable choices while planning city trips.

2. Application will nudge users towards eco-friendly transportation, accommodation, dining, and activities by integrating game elements such as points, badges, leaderboards, and challenges.

3. It could also be used to recommend places less visited or during non peak seasons through a gamified app

Feel free to suggest more options, ideas in this direction.

Required (Primary) Skills: Full-stack web/mobile dev

Project: [A3] Personalization in Conversational Recommender Systems for Tourism

Proposed Work

1. Literature Review
    - Conduct an in-depth review of existing personalization techniques in conversational systems.
    - Identify gaps and opportunities specific to conversational TRS.

2. Prototype Implementation
    - Develop a small-scale prototype to test and evaluate personalization strategies in a controlled environment.
    - Focus on integrating adaptive dialogue flows and user preference learning.

3. Synthetic Data Generation (Optional / if need be)
    - Generate synthetic datasets simulating real-world conversational interactions in tourism contexts.
    - Use these datasets to train and benchmark personalization techniques effectively.
    - Improve on our SIGIR paper (https://arxiv.org/pdf/2504.09277)

Required (Primary) Skills: Python, understanding of LLMs, data analysis, machine learning or similar

Project: [A4] Accommodation Recommendations using Booking.com open data

Proposed Work

1. Literature Review
   - improve accommodation recommendation systems by conducting a thorough analysis of the available datasets

2. Prototype Implementation
   - formulating a novel problem statement & implementation
   - feel free to combine different (relevant) datasets

Required (Primary) Skills: Python, understanding of LLMs, data analysis, machine learning or similar, creativity

Project Expectations A1-A3

- Has to be a **use-case for Tourism Recommender Systems**

- You can use any publicly available datasets or even gather your own data in addition to whatever resources are provided
    - Encouraged to use a combination of different datasets

- Build UIs, or come up with great algorithm, solve some of world's crisis — as long as it's a Tourism RS use-case, we're fine
    - If you're building an app ➡ you can either deploy it using any open (or free) platforms or submit a **.mp4** file with the screen recording of your app.
    - If you're focusing on an algorithm, make sure you have some evaluation strategies/metrics to determine how awesome your (recommendation) algo is.
    - In either case, it would be great to also have some evaluation of your app/algo either through some online/offline evaluation strategies.

- Projects with promising results will have the possibility to publish papers under the guidance of the Chair!

- Be creative! We are completely flexible!

Meeting Schedule & Plan A1-A3

Mondays from 14:00 over Teams (Link to be shared)

20.04.2026	Kick-off, organisational issues, team formation
11.05.2026	Milestone 1: Conceptualize - Introduction, motivation, problem statement, EDA
01.06.2026	Milestone 2: Experiments & Implementation 1
22.06.2026	Milestone 3: Experiments & Implementation 2

- Also, there is a possibility to schedule on-demand optional Teams discussion sessions. You will have to talk to us (drop us an email).

- Teams will present their updates in **about 15 min (plus Q&A)**

- Talks should be given out freely in English and not read out of scripts
  - Also: discussion, every participant should have comments and questions about other projects as well
  - Send slides to Ashmi (optionally)

- Attendance at your own group's project meetings is mandatory for grading
  - Attending the other group's meetings is optional (e.g., Group A must attend all Group A meetings but not Group W's)

Topics W1-W4

- Rather general topic areas
- Procedure
  - Review state-of-the-art (existing solutions) and decide on elements of solution
    - Including data and technology
  - Design and implement solution
  - Evaluation and test
    - Does not have to be extensive
    - Can be focussed on core functions
- Document everything
- Given references are meant as starting points
  - Different direction is possible

Project: [W1] Knowledge Graphs for Tourism Recommendations

- Knowledge Graphs (KGs) important approach to formally represent semantics by describing entities and their relationships
  - Useful in complex domains such as Tourism Recommender Systems (TRS)
- Goal of this project is to first construct a tourism KG, maybe based on data from DBpedia or similar sites, or review existing tourism KGs that can be used as foundation for a rec. system
  - Task is then to implemented a recommender application on this KG, with the focus on functionalities that exploit the semantic and hierarchical structure of the KG
  - Simple example is to not only suggest restaurants that fit a given type of cuisines, but also consider places that offer similar cuisines, based on the KG information
  - Recommender system and its features should be briefly tested as part of a proof-of-concept

Ref.: Hendrik, Hendrik, Sri Suning Kusumawardani, and Adhistya Erna Permanasari. "Exploring the Landscape of Tourism Knowledge Graphs: A Systematic Literature Review." 2024 9th International Conference on Information Technology and Digital Applications (ICITDA). IEEE, 2024.

Project: [W2] Decentralized Point-of-Interest Recommender System

- Usually, recommender systems are centralized, i.e. user ratings and all other information is stored and managed on centralized servers
    - Idea of decentralized recommender system has some advantages, especially in mobile scenarios -> separate personalization (i.e. management of user preferences and past behavior) from the actual recommendation
        - Users would manage their preferences, ratings, and other personal information on mobile devices to improve user control and reduce privacy concerns
        - Information can then be selectively shared with other users and server-based applications to generate personal recommendations locally on the user device
- Scenario is tourist exploring city and proactively receiving notifications about interesting points-of-interests in the vicinity
    - Local user model can then be used to generate personalized recommendations on smartphone, using global item model
- Goal of this project is to implement prototype application to put this idea into practice, including user interaction (e.g. specify which parts of their profile to share with services)

Ref.: Cai, Qiqi, et al. "Distributed Recommendation Systems: Survey and Research Directions." ACM Transactions on Information Systems 43.1 (2024): 1-38.

Project: [W3] User Control and Strategies for Influencing Algorithms in Tourism Rec.Sys.

- Modern recommender systems and other machine learning applications are often black-box systems, users - or the systems themselves - do not really know how the recommendations were generated
- Desirable to allow users some control and influence over the used algorithms
  - Also requirement of the EU Digital Services Act (DSA)
- Examples in tourism: booking sites of hotels, flights or other services provide only simple sorting options to influence rankings and do not explain details
- Goal of the topic is to investigate how users can better influence ranking and recommendation algorithms in TRS, both from a user's perspective but also algorithmic approaches
  - Users could for example use sliders to change weights of different methods in hybrid recommender system, but additional approaches are necessarily to interact with parameters of complex recommendations algorithms

Ref.: Bakalov, Fedor, et al. "An approach to controlling user models and personalization effects in recommender systems." Proceedings of the 2013 International Conference on Intelligent User Interfaces. 2013.

Project: [W4] Re-finding Personal Information in Tourism Recommender Systems

- Recommender systems are mostly recommending “new” items, i.e. items that user has not seen before
    - Personal information management (PIM) deals with finding items that the user already seen, bookmarked or otherwise interact with
- Goal of this project is to implement prototype application that supports information refinding with recommender system
    - Scenario could be tourism recommender system, with item space consisting of bookmarks, calendar entries, stored markers on maps, documents such as tickets, ...
        - Specific application scenario is flexible in this project
    - One aspect is to not only respond to explicit user request, but proactively recommend and show information to user based on current situation (location, time)
    - Another aspect is to generate and display explanations for the recommended items, because it may not be clear to user why an item is shown in certain situation

Ref.: Sappelli, Maya, Suzan Verberne, and Wessel Kraaij. "Evaluation of context-aware recommendation systems for information re-finding." Journal of the Association for Information Science and Technology 68.4 (2017): 895-910.

Milestones W1-W4

- **Online** milestone presentations on **Mondays from 16:00**
  - 1.Milestone 11.05.2026
    - Focus on what to do in more detail, technology, data etc.
  - 2.Milestone 01.06.2026
    - Focus on implementation
  - 3.Milestone 22.06.2026
    - Focus on completing prototype and evaluation/test
- Teams will present their updates in **about 15 min** (plus Q&A)
  - Talks should be given out freely in English and not read out of scripts
  - Also: discussion, every participant should have comments and questions about other projects as well
  - Send slides to Wolfgang
- Attendance at your own group's project meetings is **mandatory** for grading
  - Attending the other group's meetings is optional

Final Presentations

Mon 06.07.2026, from 14:00 (tentative): Final presentations (all students, both groups)

[online via Teams]

- Duration 30 minutes (25 mins + 5 mins Q&A) per team
- Talk should be given freely in English - not read out from scripts
- You should present slides in any electronic format i.e. PDF/.ppt
- You can use the TUM powerpoint template or your own format for the slides
  - https://portal.mytum.de/corporatedesign/index_html
- Each team member should present their contributions & challenges encountered
- **Fri 17.07.2026, 23:59 CEST: Final deadline to complete and submit documentation**

Grading & Evaluation Criteria

- Requirements for credits & grading
  - **Complete, present and document** all parts of the solution for your topic
  - Attendance in **all (online) meetings** (attendance and contribution in the milestone meetings of your group is mandatory, you can optionally joining the milestone meetings of the other group as well)
  - Grading will be based on the quality of the code and solution as presented and documented
    - Milestone and final presentations
    - Final documentation

- Submission **(Email to woerndl@in.tum.de, ashmi.banerjee@tum.de)**
  - Code in GitHub repository
  - Slides of final presentation as pdf
  - Final paper/document (see next slide)
  - Application video (.mp4 or similar) if necessary/relevant
    - Screencast showing everything the UI is capable of

Final Documentation

- **One PDF per team** with all the information on the project
    - **LaTeX is recommended** but not mandatory.
- Needs to be submitted **17.07.2026 (until 23:59 CET)**
- Guidelines
    - ACM single column template: https://www.overleaf.com/latex/templates/association-for-computing-machinery-acm-large-1-column-format-template/fsyrjmfzcwyy
    - **10-12 pages + appendix (max. 2 pages) + references (unlimited)**
    - **Table with your full names** (as on TUM transcripts), GitHub handles, and contributions should be present in the end (does not count towards the page limit)
    - Report should follow the typical structure of: title, abstract, introduction & motivation, methodology, experiments (UI screenshots if necessary), evaluation/results, conclusion & future work
    - Please add a GenAI disclosure section in the end of the report, detailing the GenAI usage for the project (does not contribute to page limit).
- **Submission through official TUM email. Please include your team# in subject. One submission per team.**

Some More Information

- Moodle
  - Moodle course to share slides and final documentation
  - Will be set up soon (also registration in TUM-Online)
- GitHub Repository
  - Please include team number in your repository name.
  - They should also have a documentation (README) file in the repo for reproducibility reasons in future.

Thank You! Time for Q&A!

Feedback Form

https://bit.ly/feedback-rs-lab